PHASE 1.3

Logical Data Model & Schema Design

Beachhead Workflow — Decision Graph Persistence Schema

Derived from the approved PRD and Domain Model (Phase 1.2)

Venture Foundry OS — Governance Review Draft

1. Database Design Principles

This schema translates the approved Domain Model (Phase 1.2) into a logical, ORM- and framework-independent persistence model. Where the Domain Model does not fully determine a storage decision — a column's exact shape, an index, a naming convention — that decision is derived from one of the seven principles below, never invented freely. Nothing here revisits the PRD or modifies the Domain Model; both are treated as frozen inputs.

- DP1 — Every table maps to a named Domain Model concept. Each logical table corresponds to exactly one Aggregate Root, child Entity, Value Object (or Value Object collection), or Relationship Type already frozen in the Domain Model. No table is added to represent a concept the Domain Model does not name.
- DP2 — Surrogate keys are a storage concern, not a business one. Per Domain Model P5, every table receives a system-generated surrogate primary key, even where a business identity is described in prose (Domain Model §14). The one exception is Regulation, whose citation is itself the natural business identity (§3, regulation) — there, the surrogate key exists only for foreign-key stability, and the citation carries a uniqueness constraint.
- DP3 — Value Object cardinality determines its storage shape. A Value Object that occurs many times per owning Entity (Discussion Entry, Alternative) becomes an owned child table. A Value Object with exactly one instance per owner (Agenda Item Reference) is embedded as columns on the owner's table. A Value Object that is the payload of a many-to-many Relationship Type (Vote, Approval) becomes an associative table keyed by that relationship's two foreign keys, not a standalone entity table.
- DP4 — Inferred fields are flagged, not silently frozen. Where the PRD or Domain Model names an object's role in the workflow but not its literal data fields (e.g., a Person's display name, a Decision's short title), this schema includes only the minimum descriptive attribute needed for that object to satisfy its stated Functional Requirement, and records every such inference in Section 14 rather than treating it as approved scope.
- DP5 — Confidentiality scoping is direct, not inherited. NFR-03 and Domain Model P6/INV-5 require that Evidence, Assumption, Risk and Obligation reuse stay within one Organization's Decision Graph. Because these Aggregate Roots reach Organization only indirectly (through whichever Decisions or Meetings currently link them), this schema carries organization_id directly on each of those four tables so the boundary is enforceable at the data layer, not only wherever application code remembers to check it.
- DP6 — Versioning and audit are a uniform pattern, not a bespoke design per table. NFR-01 (traceability) and NFR-02/INV-4 (versioning, not overwrite) apply the same way to every table they touch. This schema defines the pattern once (§8, §10) and applies it identically wherever the Domain Model's lifecycle section (§11) calls for it, rather than re-deriving column names table by table.
- DP7 — The Decision Graph is the relational model, not a separate structure. Per Domain Model P1, the graph is “a separate, read-oriented traversal structure layered on top” of the Aggregates — not a second write path. Every Relationship Type in Domain Model §9–10 resolves to a foreign key, an associative table, or a junction table (§12); there is no independent graph store in V1, per this document's own instructions.

Standard Identity & Audit Columns (present on every logical table)

Column	Description
{table}_id	System-generated surrogate primary key. Storage-only; not the business identity described in Domain Model §14 (DP2).
created_at	Timestamp the row was first persisted.
created_by_person_id	FK → person; who captured this record, per NFR-01. Omitted where another column on the same table already identifies the acting Person (e.g., decision_vote.person_id, discussion_entry.contributor_person_id, evidence.uploaded_by_person_id) — adding a second, redundant actor column would not increase traceability.

Standard Version Lineage Columns (present only where a table is marked “Versioned: Yes” in §3)

Column	Description
version_number	Integer, starting at 1 for the first-recorded row in a business-identity lineage.
supersedes_id	FK to the prior version's surrogate id, within the same table. Null on the first version.
is_current	Boolean; true for exactly one row per business-identity lineage — the row that should be read by default.
superseded_at	Timestamp a newer version replaced this one. Null while is_current = true.

Format note for Section 3: The requested subsections — Purpose, Primary business identity, Attributes, Required fields, Optional fields — are presented per table as: a Purpose sentence, a Business Identity sentence, and a single Attributes table whose third column states Required or Optional. Repeating each attribute a second and third time as separate “Required” and “Optional” lists would duplicate the same information without adding anything a reader couldn't already see in that column, so it is not done here. The two Standard Column tables above are referenced by name from each table below rather than re-listed sixteen times.

2. Schema Overview

The schema has three layers. Reference & Tenancy tables (Organization, Person, Regulation) are the objects other tables point to but that are not themselves produced by the Beachhead Workflow. Workflow Entity tables (Meeting, Decision, Evidence, Assumption, Risk, Obligation, Resolution, Action Item, Minutes) are the Aggregate Roots and child Entities that make up one Meeting → Decision Graph cycle (PRD §12). The Relationship Layer (associative and junction tables) realizes every Relationship Type from Domain Model §9–10 as either a direct foreign key, an associative table carrying a Value Object's attributes, or a plain junction table.

Layer	Purpose	Logical Tables
Reference & Tenancy	Shared or Organization-scoped data that Meetings and Decisions point to; not produced by the workflow itself.	organization, person, regulation
Workflow Entities	The Aggregate Roots and child Entities generated as one Meeting is run through the Beachhead Workflow.	meeting, decision, evidence, assumption, risk, obligation, resolution, action_item, minutes
Value Object Collections	Owned, multi-row Value Objects with no independent business identity (DP3).	decision_discussion_entry, decision_alternative
Relationship Layer	Associative tables (Value Object payload) and junction tables realizing the Domain Model's Relationship Types.	decision_vote, decision_approval, + 9 junction tables (§5)

In total this is 25 logical tables: 12 entity/generated tables, 2 owned Value Object collections, and 11 relationship-layer tables. This count maps exactly onto the twelve Core Domain Objects, six Value Objects and fifteen named Relationship Types frozen in the Domain Model — no table introduces a concept beyond that boundary.

3. Logical Tables

organization

Versioned: No · Lifecycle: single Active state (Domain Model §11)

Purpose: the tenancy and confidentiality boundary for the entire schema (Domain Model §3, P6); the customer entity whose governance is recorded (PRD §16).

Primary business identity: the distinct legal/customer entity the platform serves; identity is assigned once, at onboarding, and never reassigned (Domain Model §14).

Attribute	Description	Required / Optional
organization_id	Surrogate primary key (DP2).	Required
legal_name	The organization's registered/display name.	Required
segment	Illustrative customer segment (e.g., holding company, law firm, NGO — PRD §9 Target Segments).	Optional
jurisdiction	Country/region of registration; relevant to NFR-04 data residency.	Optional — deferred, §14
created_at	Standard audit column (§1).	Required
created_by_person_id	Not applicable at this table — onboarding precedes any Person record (§14).	Not enforced

person

Versioned: No · Lifecycle status: Active / Departed (Domain Model §11)

Purpose: an individual participant in governance — director, corporate secretary, external counsel, or auditor (PRD §16).

Primary business identity: a specific individual, scoped to one Organization in V1; identity persists through the Active → Departed transition — departure never creates a new identity (Domain Model §14, §13).

Attribute	Description	Required / Optional
person_id	Surrogate primary key.	Required
organization_id	FK → organization (MEMBER_OF).	Required
full_name	Display name used across Minutes, Resolutions and Rationale Retrieval (inferred, §14).	Required
role	PersonRole enum: Director, CorporateSecretary, ExternalCounsel, Auditor, Other (Domain Model §8).	Required
status	Active or Departed.	Required
departed_at	Populated when status = Departed.	Optional
contact_email	Not specified in the approved PRD; included as a plausible operational necessity (inferred, §14).	Optional
created_at / created_by_person_id	Standard audit columns (§1).	Required

meeting

Versioned: Yes (§8) · Lifecycle status: Open / Concluded (Domain Model §11)

Purpose: a single Board Meeting instance (PRD §16); the transactional unit for FR-01–FR-03.

Primary business identity: a specific occurrence of the board meeting for one Organization; since two meetings could share a date, identity is assigned at creation and is never derived solely from date or type (Domain Model §14).

Attribute	Description	Required / Optional
meeting_id	Surrogate primary key.	Required
organization_id	FK → organization (HELD_BY).	Required
meeting_date	Date the meeting occurred.	Required
meeting_type	MeetingType enum: Ordinary, Extraordinary, AnnualGeneral — illustrative, unconfirmed (Domain Model §8 note).	Required

Attribute	Description	Required / Optional
status	Open or Concluded.	Required
concluded_at	Set when Minutes are generated (FR-12).	Optional
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns (§8) — support post-Concluded corrections (Domain Model §11).	Required
created_at / created_by_person_id	Standard audit columns.	Required

decision

Versioned: Yes (§8) · Lifecycle status: DecisionStatus enum (Domain Model §8, §11)

Purpose: the central knowledge unit — an approved or rejected choice, together with its rationale (PRD §16, Domain Model §3).

Primary business identity: a specific decision reached (or not reached) on one agenda item at one Meeting; identity is assigned at proposal and retained through every later lifecycle state, including versioned corrections (Domain Model §14).

Attribute	Description	Required / Optional
decision_id	Surrogate primary key.	Required
meeting_id	FK → meeting (MADE_AT).	Required
agenda_item_label, agenda_item_sequence	The embedded Agenda Item Reference Value Object (Domain Model §5, DP3) — single-cardinality per Decision, so it is columns, not a child table.	Required
decision_title	Short descriptive label used when generating Minutes/Resolutions (inferred, §14).	Required
status	DecisionStatus enum: Proposed, UnderDiscussion, Voted, Approved, Rejected.	Required
rationale_text	FR-06 rationale capture. Nullable at creation; INV-2 requires it populated before status may become Approved (enforced §7, not by column nullability).	Optional (business-required pre-approval)
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns.	Required

Attribute	Description	Required / Optional
created_at / created_by_person_id	Standard audit columns.	Required

decision_discussion_entry

Versioned: No — immutability inherited from the parent Decision (DP3) · Value Object collection (Domain Model §5)

Purpose: one point raised during discussion of the agenda item, before the Decision was finalized (FR-04).

Primary business identity: none — this is a Value Object. The surrogate key below orders and addresses rows for storage purposes only (Domain Model P5, DP2/DP3); it is never referenced independently outside the context of its parent Decision.

Attribute	Description	Required / Optional
discussion_entry_id	Surrogate key (storage ordering only).	Required
decision_id	FK → decision (owning Aggregate).	Required
contributor_person_id	FK → person; satisfies NFR-01 for this row directly (§1).	Required
entry_text	The discussion point recorded.	Required
sequence_number	Ordering within the Decision's discussion.	Required
created_at	Standard audit column.	Required

decision_alternative

Versioned: No (as above) · Value Object collection (Domain Model §5)

Purpose: one option considered and not chosen, recorded alongside why it was set aside (FR-08).

Primary business identity: none — Value Object; surrogate key is storage-only (DP3).

Attribute	Description	Required / Optional
alternative_id	Surrogate key.	Required
decision_id	FK → decision.	Required
description_text	The alternative that was considered.	Required
reason_not_chosen	Why it was set aside (FR-08).	Required
sequence_number	Ordering.	Required
created_at / created_by_person_id	Standard audit columns.	Required

decision_vote

Versioned: No — immutable once cast, subject to INV-4 · Associative table realizing the Vote Value Object and the CAST_VOTE_ON Relationship Type

Purpose: a single Person's recorded choice on a Decision (FR-10).

Primary business identity: none — Value Object. The composite of decision_id + person_id is the natural key: INV-6 permits at most one Vote per Person per Decision.

Attribute	Description	Required / Optional
decision_id	FK → decision.	Required
person_id	FK → person; the voter (also satisfies NFR-01 — no separate created_by column).	Required
vote_choice	VoteChoice enum: For, Against, Abstain (Domain Model §8).	Required
cast_at	Timestamp the vote was recorded.	Required

> **Constraint cross-reference:** UNIQUE (decision_id, person_id) on this table is the schema-level enforcement of INV-6; see §7.

decision_approval

Versioned: No — immutable once recorded · Associative table realizing the Approval Value Object and the APPROVED_BY Relationship Type

Purpose: a record that a specific Person formally approved a Decision, distinct from how they voted (FR-11).

Primary business identity: none — Value Object. The composite of decision_id + person_id is treated as the natural key, constrained to at most one approval record per Person per Decision. This is an inferred constraint, not an explicit Domain Invariant — flagged in §14.

Attribute	Description	Required / Optional
decision_id	FK → decision.	Required
person_id	FK → person; the approver.	Required
approved_at	Timestamp of formal approval.	Required

minutes

Versioned: No — generated once, read-only · Value Object (Domain Model §5)

Purpose: the generated, read-only compilation of a Meeting's Discussion, Decisions, Votes and Approvals (FR-12).

Primary business identity: none — Value Object; not an independently linkable Decision Graph node (Domain Model §5 note). One row per Meeting.

Attribute	Description	Required / Optional
minutes_id	Surrogate key.	Required
meeting_id	FK → meeting, unique. A 1:1 composition, not a Relationship Type edge (§6).	Required
generated_at	When the Minutes Compilation Service (Domain Model §6) ran.	Required
compiled_content	The generated document content, or a pointer to it.	Required

resolution

Versioned: No — generated once, immutable · Entity, child of Decision (Domain Model §4)

Purpose: the formal document generated from an Approved Decision (FR-13).

Primary business identity: the specific formal document generated for one Decision, typically citable going forward by a durable resolution reference; assigned at generation and never reused for a different Decision (Domain Model §14).

Attribute	Description	Required / Optional
resolution_id	Surrogate key.	Required
decision_id	FK → decision, unique (RESULTS_IN, 1:0..1).	Required
resolution_reference	The durable citation number — the business identity described above.	Required
generated_at	When the Resolution Drafting Service ran; INV-3 gates this to decision.status = Approved.	Required
content_reference	The generated document content, or a pointer to it.	Required

action_item

Versioned: No — single Open state (Domain Model §8 note, §11) · Entity, child of Decision

Purpose: a follow-up task generated from a Decision (FR-14).

Primary business identity: a specific follow-up task arising from one Decision; identity is assigned at generation and persists regardless of later status reporting (Domain Model §14) — though V1 defines no status field to report against (see note below).

Attribute	Description	Required / Optional
action_item_id	Surrogate key.	Required

> **Deliberate omission:** No status or completion column is included. Domain Model §8 explicitly defers an ActionItemStatus enum to Future Expansion; adding one here would exceed approved V1 scope.

evidence

Versioned: Yes (§8) · Lifecycle: Active → Superseded (Domain Model §11)

Purpose: a supporting document — agenda, legal opinion, financial report — attached to a Meeting or Decision (PRD §16, FR-02/FR-03).

Primary business identity: a specific uploaded document as attached to the graph; identity is assigned at upload. Two uploads of the same underlying file are, by default, two distinct Evidence identities unless explicitly linked as the same source (Domain Model §14).

Attribute	Description	Required / Optional
evidence_id	Surrogate key.	Required
organization_id	FK → organization; direct confidentiality scoping (DP5, INV-5).	Required
evidence_type	EvidenceType enum: Agenda, SupportingDocument, LegalOpinion, FinancialReport, Other (Domain Model §8).	Required
file_reference	Pointer to the stored file.	Required
description	Display label (inferred, §14).	Optional
uploaded_by_person_id	FK → person.	Required
uploaded_at	Timestamp.	Required
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns.	Required

assumption

Versioned: Yes (§8) · Lifecycle: Active → Superseded

Purpose: a financial, legal or operational assumption underlying a Decision (PRD §16, FR-07).

Primary business identity: a specific piece of recorded reasoning; identity is assigned when first recorded and is retained even as it comes to be BASED_ON from additional later Decisions (Domain Model §14).

Attribute	Description	Required / Optional
assumption_id	Surrogate key.	Required

Attribute	Description	Required / Optional
organization_id	FK → organization; confidentiality scoping (DP5, INV-5).	Required
assumption_text	The assumption recorded.	Required
category	Descriptive grouping (financial / legal / risk, per PRD FR-07 wording) — not frozen as a Domain Model Enumeration; included as free text pending a governance decision (§14).	Optional
recorded_by_person_id / created_at	Standard audit columns.	Required
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns.	Required

risk

Versioned: Yes (§8) · Lifecycle: Active → Superseded

Purpose: a risk acknowledged or accepted as part of a Decision (PRD §16, FR-09).

Primary business identity: a specific piece of recorded reasoning; identity is assigned when first recorded and is retained as it comes to be ACCEPTED_BY additional later Decisions (Domain Model §14).

Attribute	Description	Required / Optional
risk_id	Surrogate key.	Required
organization_id	FK → organization; confidentiality scoping.	Required
risk_text	The risk recorded.	Required
recorded_by_person_id / created_at	Standard audit columns.	Required
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns.	Required

FLAG No severity, likelihood, or scoring fields are included on this table. PRD §18 explicitly places a standalone Risk Register out of V1 scope — “Risk exists in V1 only as an object linked to a Decision.” A scoring model would exceed that boundary even though it is common in risk-register schemas generally.

obligation

Versioned: Yes (§8) · Lifecycle: Active → Superseded

Purpose: a compliance or contractual obligation connected to a Decision (PRD §16, FR-09).

Primary business identity: a specific piece of recorded reasoning; identity is assigned when first recorded and is retained as it comes to be SUBJECT_TO from additional later Decisions (Domain Model §14).

Attribute	Description	Required / Optional
obligation_id	Surrogate key.	Required
organization_id	FK → organization; confidentiality scoping.	Required
obligation_text	The obligation recorded.	Required
regulation_id	FK → regulation (DERIVED_FROM). Modeled NOT NULL per the Domain Model's many:1 cardinality — see the flagged tension in §14.	Required (flagged)
recorded_by_person_id / created_at	Standard audit columns.	Required
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns.	Required

regulation

Versioned: Yes (§8) · Lifecycle: Active → Superseded · Shared reference data, not Organization-scoped (Domain Model §2, §13)

Purpose: a regulatory reference relevant to a Decision or Organization (PRD §16).

Primary business identity: unlike every other Entity in this schema, the regulatory citation itself (e.g., a named Act and section) is the natural business identity — two references to the same citation are the same Regulation record by definition (Domain Model §14). This is enforced as a uniqueness constraint on citation, layered over the surrogate primary key kept for FK stability (DP2).

Attribute	Description	Required / Optional
regulation_id	Surrogate primary key.	Required
citation	The natural/business key described above; unique.	Required
title	Descriptive title of the regulation.	Required
jurisdiction	The applicable jurisdiction.	Optional
source_reference	Link or pointer to the regulatory text.	Optional
created_by_person_id	May not map to a workflow Person — Regulation is centrally curated (Domain Model §13); nullable (§14).	Optional
created_at	Standard audit column.	Required

Attribute	Description	Required / Optional
version_number, supersedes_id, is_current, superseded_at	Standard version lineage columns.	Required

4. Primary and Foreign Key Relationships

Every table's primary key is its own {table}_id surrogate column (DP2); this is not repeated below. The table lists only foreign keys — what each table points to, and why.

Table	Foreign Key(s) → Referenced Table	Relationship / Note
organization	—	Root of the tenancy boundary; no outbound FKs.
person	organization_id → organization	MEMBER_OF.
meeting	organization_id → organization; created_by_person_id → person; supersedes_id → meeting	HELD_BY; version lineage self-reference.
decision	meeting_id → meeting; created_by_person_id → person; supersedes_id → decision	MADE_AT; version lineage self-reference.
decision_discussion_entry	decision_id → decision; contributor_person_id → person	Owned VO collection (DP3).
decision_alternative	decision_id → decision; created_by_person_id → person	Owned VO collection (DP3).
decision_vote	decision_id → decision; person_id → person	CAST_VOTE_ON.
decision_approval	decision_id → decision; person_id → person	APPROVED_BY.
minutes	meeting_id → meeting (unique)	1:1 VO composition, not a Relationship Type.
resolution	decision_id → decision (unique)	RESULTS_IN, 1:0..1.
action_item	decision_id → decision; assigned_to_person_id → person	RESULTS_IN, 1:many; ASSIGNED_TO.
evidence	organization_id → organization; uploaded_by_person_id → person; supersedes_id → evidence	Confidentiality scoping (DP5); version lineage.
assumption	organization_id → organization; recorded_by_person_id → person; supersedes_id → assumption	Confidentiality scoping; version lineage.

Table	Foreign Key(s) → Referenced Table	Relationship / Note
risk	organization_id → organization; recorded_by_person_id → person; supersedes_id → risk	Confidentiality scoping; version lineage.
obligation	organization_id → organization; regulation_id → regulation; recorded_by_person_id → person; supersedes_id → obligation	Confidentiality scoping; DERIVED_FROM (flagged, §14); version lineage.
regulation	created_by_person_id → person (nullable); supersedes_id → regulation	Shared reference data; version lineage.
meeting_attendee	meeting_id → meeting; person_id → person	ATTENDED.
meeting_evidence_link	meeting_id → meeting; evidence_id → evidence	SUPPORTED_BY.
decision_evidence_link	decision_id → decision; evidence_id → evidence	SUPPORTED_BY.
decision_assumption_link	decision_id → decision; assumption_id → assumption	BASED_ON.
decision_risk_link	decision_id → decision; risk_id → risk	ACCEPTS.
decision_obligation_link	decision_id → decision; obligation_id → obligation	SUBJECT_TO.
organization_regulation_link	organization_id → organization; regulation_id → regulation	SUBJECT_TO.
decision_regulation_link	decision_id → decision; regulation_id → regulation	GOVERNED_BY.
decision_informed_by_link	decision_id → decision; informing_decision_id → decision	INFORMED_BY — self-referential.

5. Junction Tables

The nine tables below are pure junctions — they resolve a many-to-many Relationship Type but carry no Value Object attributes of their own, unlike decision_vote and decision_approval (§3), which are associative tables carrying the Vote and Approval Value Objects respectively and are therefore documented there instead of here (DP3).

Junction Table	Realizes Relationship Type	Columns	Cardinality
meeting_attendee	Person ATTENDED Meeting	meeting_id, person_id, created_at	many:many

Junction Table	Realizes Relationship Type	Columns	Cardinality
meeting_evidence_link	Meeting SUPPORTED_BY Evidence	meeting_id, evidence_id, created_at, created_by_person_id	many:many
decision_evidence_link	Decision SUPPORTED_BY Evidence	decision_id, evidence_id, created_at, created_by_person_id	many:many
decision_assumption_link	Decision BASED_ON Assumption	decision_id, assumption_id, created_at, created_by_person_id	many:many
decision_risk_link	Decision ACCEPTS Risk	decision_id, risk_id, created_at, created_by_person_id	many:many
decision_obligation_link	Decision SUBJECT_TO Obligation	decision_id, obligation_id, created_at, created_by_person_id	many:many
organization_regulation_link	Organization SUBJECT_TO Regulation	organization_id, regulation_id, created_at	many:many
decision_regulation_link	Decision GOVERNED_BY Regulation	decision_id, regulation_id, created_at, created_by_person_id	many:many
decision_informed_by_link	Decision INFORMED_BY Decision (self-referential)	decision_id, informing_decision_id, created_at, created_by_person_id	many:many

> **Composite keys:** Each junction table's primary key is the composite of its two foreign key columns (e.g., (meeting_id, person_id) on meeting_attendee); no separate surrogate key is introduced for a pure junction, since the pair itself is exactly the fact being recorded.

6. Cardinality Matrix

This mirrors the Domain Model's Relationship map (§9) with one added column: how each edge is actually stored. The final three rows are Value Object compositions (DP3), not Relationship Type edges in the Domain Model's own sense (Domain Model §5 note on Minutes) — included here for completeness of the storage picture.

From	Relationship	To	Cardinality	Storage Mechanism
Meeting	HELD_BY	Organization	many:1	Direct FK — meeting.organization_id
Person	MEMBER_OF	Organization	many:1	Direct FK — person.organization_id
Person	ATTENDED	Meeting	many:many	Junction — meeting_attendee
Decision	MADE_AT	Meeting	many:1	Direct FK — decision.meeting_id
Meeting	SUPPORTED_BY	Evidence	many:many	Junction — meeting_evidence_link

From	Relationship	To	Cardinality	Storage Mechanism
Decision	SUPPORTED_BY	Evidence	many:many	Junction — decision_evidence_link
Decision	BASED_ON	Assumption	many:many	Junction — decision_assumption_link
Decision	ACCEPTS	Risk	many:many	Junction — decision_risk_link
Decision	SUBJECT_TO	Obligation	many:many	Junction — decision_obligation_link
Organization	SUBJECT_TO	Regulation	many:many	Junction — organization_regulation_link
Decision	GOVERNED_BY	Regulation	many:many	Junction — decision_regulation_link
Obligation	DERIVED_FROM	Regulation	many:1	Direct FK — obligation.regulation_id (flagged, §14)
Person	CAST_VOTE_ON	Decision	many:many	Associative table (VO) — decision_vote
Decision	APPROVED_BY	Person	many:many	Associative table (VO) — decision_approval
Decision	RESULTS_IN	Resolution	1 : 0..1	Direct FK, unique — resolution.decision_id
Decision	RESULTS_IN	Action Item	1 : many	Direct FK — action_item.decision_id
Action Item	ASSIGNED_TO	Person	many:1	Direct FK — action_item.assigned_to_person_id
Decision	INFORMED_BY	Decision	many:many, self-ref.	Junction — decision_informed_by_link
Meeting	(composes)	Minutes	1:1	Direct FK, unique — minutes.meeting_id — VO composition, not a graph edge
Decision	(composes)	Discussion Entry	1:many	Owned child table — decision_discussion_entry — VO composition
Decision	(composes)	Alternative	1:many	Owned child table — decision_alternative — VO composition

7. Constraints

Uniqueness

- UNIQUE (decision_id, person_id) on decision_vote — direct enforcement of INV-6.
- UNIQUE (decision_id, person_id) on decision_approval — inferred integrity rule, not an explicit Domain Invariant (§14).
- UNIQUE decision_id on resolution — Decision RESULTS_IN Resolution is 1:0..1.
- UNIQUE meeting_id on minutes — a 1:1 composition.
- UNIQUE citation on regulation — the natural business key (Domain Model §14).
- UNIQUE (decision_id, informing_decision_id) on decision_informed_by_link, plus a check that informing_decision_id ≠ decision_id — a Decision cannot inform itself.
- Within any version lineage (§8), exactly one row has is_current = true. This spans a group of rows, not a single row, so it cannot be a simple column constraint — see the Business Constraints table below.

Referential Integrity

- All foreign keys default to RESTRICT on delete, consistent with the schema's no-hard-delete stance (§9) and INV-10.
- organization_id on evidence, assumption, risk and obligation must match the organization_id reachable through every Decision or Meeting that links to them (INV-5). This is a cross-table condition and is enforced at the application/trigger layer, not by the foreign key alone.
- Where supersedes_id is present, it must reference a prior row of the same business-identity lineage in the same table — also an application/trigger-layer check, since a foreign key alone cannot compare business identity across rows.

Business Constraints Derived from Domain Invariants

Invariant	Rule (abbreviated)	Enforcement Mechanism	Layer
INV-1	A Decision is MADE_AT exactly one Meeting.	meeting_id NOT NULL FK on decision.	Relational
INV-2	A Decision cannot become Approved without Rationale, ≥1 Vote, and ≥1 Approval.	Check rationale_text IS NOT NULL and matching rows exist in decision_vote / decision_approval before the status transition is allowed.	Application / Trigger
INV-3	A Resolution can only be generated for an Approved Decision.	Check decision.status = Approved before insert into resolution.	Application / Trigger
INV-4	Post-terminal fields are immutable; corrections are new versions.	is_current / supersedes_id lineage (§8) plus a rule blocking in-place edits to non-current or terminal rows.	Relational (lineage columns) + Application (edit rule)

Invariant	Rule (abbreviated)	Enforcement Mechanism	Layer
INV-5	Evidence/Assumption/Risk/Obligation reuse stays within one Organization.	organization_id carried directly on all four tables (DP5), cross-checked against every linking Decision/Meeting's organization_id.	Relational (FK) + Application (cross-check)
INV-6	At most one Vote per Person per Decision.	UNIQUE (decision_id, person_id) on decision_vote.	Relational
INV-7	An Action Item has exactly one Decision and exactly one owning Person.	decision_id and assigned_to_person_id both NOT NULL on action_item.	Relational
INV-8	A Meeting needs ≥1 attendee before a Decision can be created against it.	Check a matching row exists in meeting_attendee before insert into decision.	Application / Trigger
INV-9	Every Relationship connects two existing, non-superseded Entities of a permitted type pair.	FK existence checks; is_current checks; the fixed junction-table set itself only allows the type pairs the Domain Model defines.	Relational (structure) + Application (non-superseded check)
INV-10	No permanent deletion of Approved-linked records.	No DELETE operation is modeled for any versioned entity or its linked Resolutions/Meetings (§9).	Application / Procedural

8. Versioning Strategy

NFR-02 and INV-4 require that once a Decision (and, per Domain Model §11, Meeting, Evidence, Assumption, Risk, Obligation and Regulation) reaches a terminal or Superseded-eligible state, corrections are recorded as new versions rather than overwrites. This schema implements that with a single pattern — version_number, supersedes_id, is_current, superseded_at (§1) — applied identically to those seven tables, and to no others.

A subtlety worth stating plainly: the surrogate primary key of a versioned table (e.g., decision_id) identifies one row — one version — not the enduring real-world Decision described in Domain Model §14. The business identity is operationally represented by the chain of rows linked through supersedes_id, all sharing the same origin row. Application and retrieval code should default to querying is_current = true, and should traverse supersedes_id only when full history — an audit trail or a “what did we believe when we approved this” reconstruction — is explicitly needed.

Resolution, Action Item and Minutes are deliberately excluded from this pattern: the Domain Model models each as generated once and immutable, with no correction path defined for V1 (§11). If a future requirement introduces the ability to correct a Resolution, that is new lifecycle behavior requiring founder approval under the Decision Freeze — not something this schema should pre-build.

9. Soft Delete / Historical Record Strategy

No table in this schema defines a DELETE operation for a row that has become part of the historical record. INV-10 requires this for Approved Decisions, Resolutions and Meetings specifically; this schema applies the same no-hard-delete stance uniformly, in keeping with NFR-06's right-sized-architecture guidance — one deletion policy is simpler to reason about and to audit than a per-table exception list.

A generic deleted_at flag is deliberately not introduced. The Domain Model's own lifecycle states already do that job wherever the Domain Model defines one: is_current = false marks a Superseded version of Decision, Meeting, Evidence, Assumption, Risk, Obligation or Regulation (§8); status = Departed marks a Person who has left. Adding a second, parallel "soft delete" column on top of those would duplicate the same fact through two mechanisms.

> **Tables with no deactivation mechanism:** Organization, Resolution, Action Item, Minutes, the two Value Object collections, and every junction/associative table have no modeled non-active state at all — the Domain Model defines none for them (§11 covers only Decision, Meeting, Person, Organization, and the five reusable reference Entities). This is not an oversight in this schema; it is a faithful mirror of a boundary the Domain Model itself did not extend past. If a future need arises to deactivate, say, an Organization, that is Future Expansion territory (§14).

One exception applies before a Decision reaches a terminal state: junction rows linking a still-Proposed or UnderDiscussion Decision to Evidence, Assumption, Risk, Obligation or Regulation may be hard-deleted, since they are not yet part of an immutable historical record. Once the parent Decision becomes Approved or Rejected, INV-4 prohibits removing them — this is a status-conditional rule enforced at the application layer, not a difference in table structure.

10. Audit & Traceability Requirements

NFR-01 requires that "every Decision Graph entry retains an immutable record of who captured it and when." Because Relationship instances are themselves Decision Graph entries (FR-15), this schema carries created_at and an acting-Person column on every table in §3–§5, including the junction tables — not only on the Entity tables.

Combined with the version lineage in §8, this produces a complete audit trail without a separate audit-log table: for a versioned Entity, each correction is a new row with its own created_at / created_by_person_id, and the row it replaced is retained untouched rather than overwritten. The version lineage is the audit log for those seven tables. For every other table — Person, Organization, the Value Object collections, and the junction/associative tables — the single created_at / acting-Person pair is the complete traceability record, since those rows are either immutable by convention or governed entirely by their parent Decision's own immutability (INV-4).

11. Indexing Recommendations (logical only)

These are logical recommendations — concepts any relational engine can express — not a specific product's physical index syntax.

- organization_id on person, meeting, evidence, assumption, risk and obligation. Because NFR-03 scopes nearly every query to one Organization, this is the single most load-bearing index family in the schema.
- decision.status — supports the Decision Approval Eligibility Service and any “decisions awaiting approval” view.
- decision.meeting_id paired with decision.is_current — the common “current decisions for this meeting” traversal.
- decision_vote(decision_id) and decision_vote(person_id) — supports tallying one Decision's votes and reading one Person's voting history, the two inverse directions named in the Domain Model's Relationship Semantics (§10).
- A filtered/partial index concept — an index limited to rows where is_current = true — on every versioned table (§8). The large majority of queries want only the current version; version history is a minority-access path.
- meeting.meeting_date — the chronological listing a Corporate Secretary defaults to.
- A text-search-capable index on decision.rationale_text, decision_discussion_entry.entry_text and evidence.description — directly supports FR-16 Rationale Retrieval and FR-18 AI Retrieval. Left as a logical requirement only: whether the concrete mechanism is a full-text index or a semantic/vector index is a System Architecture decision, out of scope here.
- regulation.citation — already unique (§7); also the primary lookup path for shared reference data.

12. Graph Representation Strategy

How the Decision Graph Is Represented Logically

Per this document's own instructions, the relational model is the canonical, persisted representation of the Decision Graph in V1 — there is no separate graph database. This is also the more faithful reading of the Domain Model itself: Principle P1 describes the graph as “a separate, read-oriented traversal structure layered on top” of the Aggregates, not a second write path. Every Relationship Type in Domain Model §9–10 resolves to exactly one of the three mechanisms catalogued in §6 — a direct foreign key, an associative table carrying a Value Object, or a junction table.

Relationship Storage

Because the object types and edges are closed — Domain Model P3 freezes the twelve Core Domain Objects, and §9–10 freeze the named Relationship Types — the “graph” that FR-16/FR-18 traverse is not a special data structure. It is the same relational tables in §3–§5, walked via a fixed, known set of joins rather than an open-ended, arbitrary-schema graph engine.

Traversal Considerations

The Rationale Retrieval Service (Domain Model §6) retrieving “the full connected rationale” for one Decision means a bounded fan-out from a single decision row: decision_discussion_entry, decision_alternative, decision_vote, decision_approval, and the four link tables to Assumption, Risk, Obligation and Evidence, plus decision_regulation_link. This is a fixed join list, not an open-ended graph walk, which is precisely what lets the relational model serve FR-16/FR-18 without a dedicated graph engine — the fan-out is knowable in advance because the object model is frozen.

One relationship is a genuine exception: Decision INFORMED_BY Decision (self-referential, many:many via decision_informed_by_link) is the sole multi-hop relationship in the model. Answering “which earlier decisions informed this one, and which informed those” requires a recursive traversal — a general relational capability (e.g., a recursive query), named here only as a logical requirement, not a specific engine feature. This directly serves the PRD's “Decision Graph reuse” success metric (PRD §8), which is explicitly a multi-hop reuse count.

For AI Retrieval performance (FR-18), an optional read-side addition is worth naming here even though it introduces no new entity: a fully-derived “rationale bundle” cache, pre-joining the fan-out above per current-version Decision. This must remain strictly derived — regenerable from the relational tables at any time, with no information that exists only in the cache — so it does not become a second source of truth alongside the tables in §3. See §14.

13. Mapping Matrix (Domain Model → Tables)

Domain Model Object	Classification	Logical Table(s)
Decision	Aggregate Root / Entity	decision
Meeting	Aggregate Root / Entity	meeting
Resolution	Entity (child of Decision)	resolution
Person	Aggregate Root / Entity	person
Organization	Aggregate Root / Entity	organization
Regulation	Aggregate Root / Entity	regulation
Risk	Aggregate Root / Entity	risk
Assumption	Aggregate Root / Entity	assumption
Obligation	Aggregate Root / Entity	obligation
Evidence	Aggregate Root / Entity	evidence
Action Item	Entity (child of Decision)	action_item
Relationship	Specialized edge types	Realized as FKs / associative tables / junction tables — see §4–§6; no single table

Domain Model Object	Classification	Logical Table(s)
Discussion Entry	Value Object (collection)	decision_discussion_entry
Agenda Item Reference	Value Object (single)	decision.agenda_item_label, decision.agenda_item_sequence
Alternative	Value Object (collection)	decision_alternative
Vote	Value Object (edge attribute)	decision_vote
Approval	Value Object (edge attribute)	decision_approval
Minutes	Value Object	minutes
All 18 Relationship instances (≈ 15 named types)	Relationship Types	See §6 Cardinality Matrix for the complete From / To / Storage mapping

14. Assumptions and Deferred Decisions

Every item below was flagged inline as it arose, in keeping with the pattern already established by the PRD and Domain Model's own Modeling Notes. None represents a silent addition to approved scope; each is either a minimal, necessary persistence detail or an open question surfaced for governance rather than resolved unilaterally here.

#	Assumption / Deferred Item	Treatment in This Schema
1	Descriptive/display fields not explicit in the PRD.	person.full_name, person.contact_email, meeting.meeting_type labels, decision.decision_title, action_item.due_date, evidence.description and organization.segment/jurisdiction are included as minimal necessary fields, each marked Optional or explicitly flagged inline in §3, not treated as approved business scope.
2	assumption.category (financial / legal / risk).	PRD FR-07 names these descriptively, but the Domain Model did not freeze an Enumeration for them (§8). Included as an optional free-text column pending a governance decision on whether to formalize it as an enum.
3	Risk scoring (severity, likelihood).	Deliberately excluded. PRD §18 places a standalone Risk Register out of V1 scope; Risk exists only as a Decision-linked object, so no scoring model is introduced.

#	Assumption / Deferred Item	Treatment in This Schema
4	obligation.regulation_id modeled NOT NULL.	This follows the Domain Model's stated many:1 DERIVED_FROM cardinality (§9). However, the PRD's own Core Domain Object definition of Obligation (§16) also describes "contractual" obligations, which need not derive from a Regulation. This tension is inherited from the approved Domain Model and is carried forward rather than resolved here; it should be clarified at governance review before implementation.
5	meeting.meeting_type enum values.	Already flagged as illustrative/unconfirmed in the Domain Model (§8); carried forward unchanged, not re-litigated here.
6	decision_approval per-Person uniqueness.	Enforced as at most one approval record per (decision_id, person_id), by analogy with INV-6's Vote rule. This is an inferred integrity rule, not an explicit Domain Invariant.
7	Data residency / jurisdiction (NFR-04).	organization.jurisdiction is included as an optional, nullable field. No region-specific storage or hosting behavior is encoded; NFR-04 itself states this requires confirmation before production launch.
8	Regulation curation process.	Regulation is modeled as a single, non-Organization-scoped table (Domain Model §2, §13). Who maintains it and how updates are versioned as a workflow is not specified in the PRD or Domain Model and is out of scope for this document.
9	Read-side "rationale bundle" cache for AI retrieval (§12).	Proposed only as an optional, fully-derived, regenerable cache. It is not part of the canonical schema and introduces no new entity or source of truth.
10	Numeric success-metric targets, SLAs, uptime, and confirmed data-residency jurisdiction.	These remain open Potential Strategic Changes carried forward from PRD §8 and §14. This schema adds only the instrumentation-supporting columns (timestamps, version lineage) needed to eventually measure them, without asserting target values.

Governance Note

This completes the Phase 1.3 deliverable. This document derives entirely from the approved PRD and the approved Domain Model; neither was revisited or modified, per the operating instructions for this phase. Ten items were made explicit inline as Assumptions and Deferred Decisions (Section 14) rather than silently decided, the most consequential being the inherited DERIVED_FROM cardinality tension on Obligation (Item 4) and the deliberate exclusion of Risk scoring fields (Item 3). Per Venture Foundry OS operating instructions, this document now stops and awaits governance approval before proceeding to System Architecture.